Literature Review Draft

Explainable Transformer-Based Analysis of Telugu-English Code-Mixed News Comments

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2. Related Work

2.1 Code-Mixed NLP: Challenges and General Approaches

Code-mixing — the practice of alternating between two or more languages within a single utterance — is pervasive in Indian social media discourse. Early work on code-mixed NLP focused predominantly on Hindi-English (Hinglish), establishing foundational tasks such as language identification, Part-of-Speech (POS) tagging, and sentiment analysis on noisy social media text [SemEval 2020 Task 9, Patwa et al., 2020]. These efforts revealed that standard monolingual and even multilingual models fail to adequately capture the linguistic patterns of code-switched text, primarily due to script mixing, informal orthography, and the absence of standardized spelling conventions.

The challenge is compounded for Dravidian languages, where the native script (e.g., Telugu Unicode) frequently co-occurs with Romanized transliterations in the same sentence — a phenomenon sometimes called Tenglish (Telugu-English) or Tenglish code-switching. Such mixed-script text defeats standard tokenizers and embedding models trained on clean corpora, motivating the need for specialized preprocessing and model selection.

2.2 Transformer Models for Indian Language NLP

The release of multilingual BERT (mBERT; Devlin et al., 2019) marked a turning point for low-resource NLP, offering a single pretrained model covering 104 languages including Telugu. However, mBERT's vocabulary allocation to Indian languages is disproportionately small relative to their speaker populations, limiting its effectiveness.

XLM-R (Conneau et al., 2020) improved upon mBERT through training on a larger multilingual corpus (CC-100), yielding stronger cross-lingual transfer, particularly for low-resource settings. Despite this, XLM-R was not explicitly optimized for Indian language code-mixing.

MuRIL (Khanuja et al., 2021) directly addressed this gap. Developed at Google India, MuRIL was trained on 17 Indian languages plus their transliterated variants, making it uniquely suited to handle both native-script and Romanized text simultaneously — precisely the kind of mixed-script input found in Tenglish news comments. MuRIL consistently outperforms mBERT on Indian language benchmarks and has demonstrated strong results in code-mixed classification tasks [Khanuja et al., 2021].

IndicBERT (Kakwani et al., 2020), released as part of the AI4Bharat IndicNLP suite, offers another Indian-language-specific alternative, pretrained on a large curated corpus of 12 Indic languages including Telugu. While IndicBERT was not designed specifically for code-mixing, its strong Telugu monolingual pretraining makes it a relevant baseline for our task.

2.3 Telugu-English Code-Mixed Text: Specific Prior Work

Research specifically targeting Telugu-English code-mixed text remains sparse relative to Hindi-English or Tamil-English work, representing the primary gap this paper addresses.

Kusampudi et al. (2021) performed sentiment analysis on Telugu-English code-mixed data with unsupervised normalization, demonstrating that preprocessing quality significantly affects downstream classification. Their work used the CMTE-IIITH dataset but did not extend to toxicity or fake news detection, nor did it incorporate explainability.

Rayala et al. (2023) proposed sentiment analysis of Tenglish data using syllable and word embeddings, reporting a 6% accuracy gain over baseline transformer models on the CMTE-IIITH and CMTE-NITANP datasets. This work confirms the benefit of embedding strategies tailored to Telugu morphology but again stops at sentiment — neither toxicity detection nor fake news identification was explored.

Kodirekka and Srinagesh (2022, 2023) investigated aspect-based sentiment analysis and preprocessing pipelines for Telugu-English code-mixed text, providing useful insights into handling transliteration normalization for Tenglish data. Their transliteration pipeline informed our own preprocessing design (Section 4).

The DravidianLangTech shared tasks (2021–2024), co-located with EACL, have provided the most directly relevant benchmarks for offensive and hate speech detection in Dravidian code-mixed languages. The HOLD-Telugu shared task at DravidianLangTech 2024 (Premjith B. et al., 2024) specifically targeted hate and offensive language detection in Telugu code-mixed text, with participating systems employing mBERT, XLM-R, and IndicBERT-based models. The best performing system achieved a macro F1 of 0.70 using fine-tuned Indic-SBERT [CUET_Binary_Hackers, 2024]. Crucially, none of the submissions incorporated explainability methods, and no prior work in this shared task series has addressed fake news detection jointly with toxicity detection on a YouTube comment corpus.

2.4 Fake News Detection in Low-Resource Languages

Fake news detection has received growing attention in the Indian NLP community, though work on Telugu specifically is limited. Thaokar et al. (2022) proposed a multi-linguistic fake news detector covering Hindi, Marathi, and Telugu, using machine learning classifiers on news article text — a fundamentally different domain from user-generated social media comments. Raja et al. (2022, 2023) applied transformer-based transfer learning for fake news detection in Dravidian languages including Telugu, achieving competitive results on monolingual news article corpora.

Critically, no prior work has combined fake news detection with toxicity detection on Telugu-English code-mixed YouTube comment data. The closest analogous work exists for other low-resource language families: Raja et al. (2023) addressed Malayalam fake news using XLM-R at DravidianLangTech, and De et al. (2022) proposed a multilingual transformer approach to fake news detection in several low-resource languages — but Telugu code-mixed comments were absent from both.

2.5 Explainability in Transformer-Based NLP

Explainability for NLP models has become increasingly important as transformer-based systems are deployed in high-stakes domains. SHAP (SHapley Additive exPlanations; Lundberg and Lee, 2017) and LIME (Local Interpretable Model-agnostic Explanations; Ribeiro et al., 2016) are the two dominant post-hoc explanation frameworks, providing token-level attribution scores that indicate which words drove a model's decision.

In the context of hate speech and toxicity detection, explainability methods have been applied to English and Hindi datasets to reveal model reliance on surface-level markers [HateExplain, Mathew et al., 2021]. However, explainability analysis for Telugu code-mixed text — where the model must navigate both Telugu script tokens and Romanized Telugu words simultaneously — has not been studied. This is a meaningful gap, as it remains unknown whether models like MuRIL attend to Telugu-script tokens, Roman-script tokens, or mixed-script phrases when making toxicity and fake news classifications.

2.6 Research Gap (Summary)

The survey above reveals four specific gaps that this paper addresses:

1. **Task combination**: No prior work jointly models toxicity detection and fake news detection on Telugu code-mixed text. Existing work treats these as separate tasks on separate datasets.
2. **Data source**: Prior Telugu NLP datasets draw from Twitter or curated news articles. No dataset of Telugu-English code-mixed YouTube news comments exists.
3. **Explainability**: DravidianLangTech shared tasks and related work do not incorporate SHAP or LIME analysis. The interpretability of transformer decisions on mixed-script Tenglish text is unexplored.
4. **Model comparison on Tenglish**: While mBERT, XLM-R, IndicBERT, and MuRIL have been applied individually to Telugu tasks, no study provides a systematic multi-model comparison on Tenglish news comment data across dual classification tasks.

This paper directly fills these gaps through: (1) a new YouTube-sourced Tenglish comment dataset with dual toxicity and fake news labels, (2) a transliteration normalization pipeline for mixed-script preprocessing, (3) a systematic comparison of four transformer baselines, and (4) SHAP- and LIME-based explainability analysis of model decisions on Tenglish text.

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